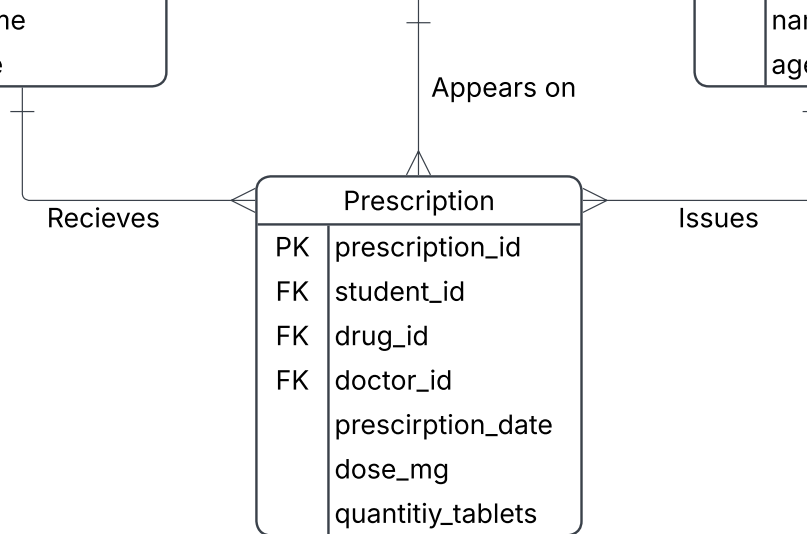


Student	
PK	student_id
	program_name
	study_year
	name
	age

StudyDrug	
PK	drug_id
	drug_producer
	drug_name

Doctor	
PK	doctor_id
	role
	department
	name
	age



View/Query results

StudentReport	
PK/FK	student_id
	amount_of_perscriptions_received
	average_doses
	average_amount

DoctorReport	
PK/FK	doctor_id
	amount_of_prescriptions_written
	average_doses
	average_amount

Database normalization

1NF Atomic values

Before

student_id	name	prescription_no	drug_id	prescription_date
S01	Alex	1, 2	D01, D02	2026-09-01, 2026-09-03
S02	Sam	1	D01	2026-09-04
S03	Jamie	1	D02	2026-09-05

After

student_id	name	prescription_no	drug_id	prescription_date
S01	Alex	1	D01	2026-09-01
S01	Alex	2	D02	2026-09-03
S02	Sam	1	D01	2026-09-04
S03	Jamie	1	D02	2026-09-05

Before prescription numbers, drug IDs and dates are stored as lists. These cells violate 1NF, and atomicity principle.

After normalization each prescription becomes a separate row. The matching number, drug and date stay together. Each prescription contains one drug.

Each cell now contains one value, so individual prescriptions can be searched and counted. Student details still repeat. The next step removes that duplication.

The primary key is (student_id, prescription_no). Prescription numbers restart for each student, so both columns are needed to identify a row.

2NF Remove partial dependencies

Before

student_id	prescription_no	name	program_name	drug_id
S01	1	Alex	Psychology	D01
S01	2	Alex	Psychology	D02
S02	1	Sam	Economics	D01
S03	1	Jamie	Biology	D02

After Student

student_id PK	name	date_of_birth	program_name	study_year
S01	Alex	2003-04-12	Psychology	2
S02	Sam	2002-11-08	Economics	3
S03	Jamie	2004-02-20	Biology	1

After Prescription

student_id PK FK	prescription_no PK	drug_id	doctor_id	prescription_date
S01	1	D01	DR01	2026-09-01
S01	2	D02	DR01	2026-09-03
S02	1	D01	DR02	2026-09-04
S03	1	D02	DR02	2026-09-05

Before the name and program_name depend only on student_id, not the whole key (student_id, prescription_no). This is a partial dependency.

Now student details move to Student. Prescription keeps student_id as a foreign key.

Each student fact is stored once. Updating a student name no longer requires changing several prescription rows, and students can exist without prescriptions.

Tables show relevant columns only. Dose, quantity and the drug and doctor descriptions remain in the prescription table until the next step.

3NF Remove transitive dependencies

Before

prescription_id	drug_id	drug_name	doctor_id	doctor_name
P001	D01	StudyTab A	DR01	Dr Lee
P002	D02	StudyTab B	DR01	Dr Lee
P003	D01	StudyTab A	DR02	Dr Noor
P004	D02	StudyTab B	DR02	Dr Noor

After StudyDrug

drug_id PK	drug_name	drug_producer
D01	StudyTab A	Producer A
D02	StudyTab B	Producer B

After Doctor

doctor_id PK	name	role	department
DR01	Dr Lee	Physician	Student Health
DR02	Dr Noor	Physician	General Practice

After Prescription

prescription_id PK	student_id FK	drug_id FK	doctor_id FK	prescription_date	dose_mg	quantity_tablets
P001	S01	D01	DR01	2026-09-01	5	20
P002	S01	D02	DR01	2026-09-03	10	15
P003	S02	D01	DR02	2026-09-04	5	10
P004	S03	D02	DR02	2026-09-05	10	20

Currently, drug_name depends on drug_id, and doctor_name depends on doctor_id. Storing these descriptions in Prescription creates transitive dependencies.

After the change drug and doctor details move to separate tables. Prescription retains their IDs and its event details. A global prescription_id replaces the student-local key. This key choice is separate from normalization.

Drug and doctor descriptions can now be updated once. Deleting a prescription does not delete their details. Student remains as shown in 2NF. Under the stated dependencies, the final four tables meet 3NF.